mlwork

August 5, 2025

[ ]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings

warnings.filterwarnings('ignore')
```

[ ]:
```
df = pd.read_csv('/insurance.csv')
```

[ ]:
```
df
```

[ ]:
```
      age     sex     bmi  children smoker     region      charges
0      19  female  27.900         0    yes  southwest  16884.92400
1      18    male  33.770         1     no  southeast   1725.55230
2      28    male  33.000         3     no  southeast   4449.46200
3      33    male  22.705         0     no  northwest  21984.47061
4      32    male  28.880         0     no  northwest   3866.85520
…     …       …       …         …      …          …            …
1333   50    male  30.970         3     no  northwest  10600.54830
1334   18  female  31.920         0     no  northeast   2205.98080
1335   18  female  36.850         0     no  southeast   1629.83350
1336   21  female  25.800         0     no  southwest   2007.94500
1337   61  female  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]
```

1 EDA

[ ]:
```
df.shape
```

[ ]:
```
(1338, 7)
```

[ ]:
```
df.head(7)
```

[ ]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
5	31	female	25.740	0	no	southeast	3756.62160
6	46	female	33.440	1	no	southeast	8240.58960

[ ]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

[ ]:

```
df.describe()
```

[ ]:

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

[ ]:

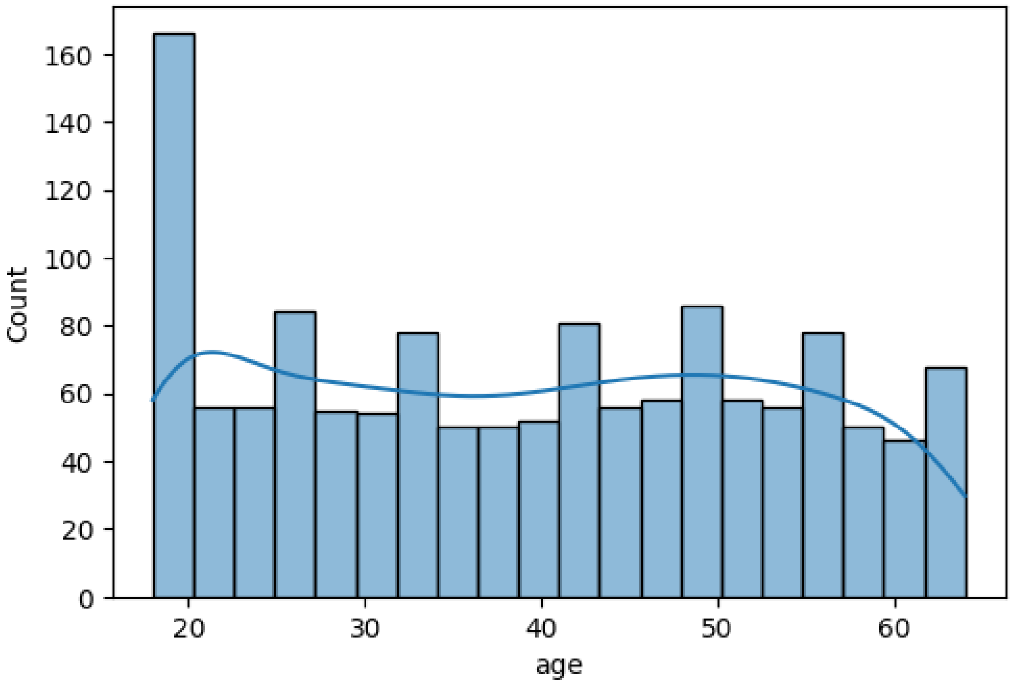
```
df.isnull().sum()
```

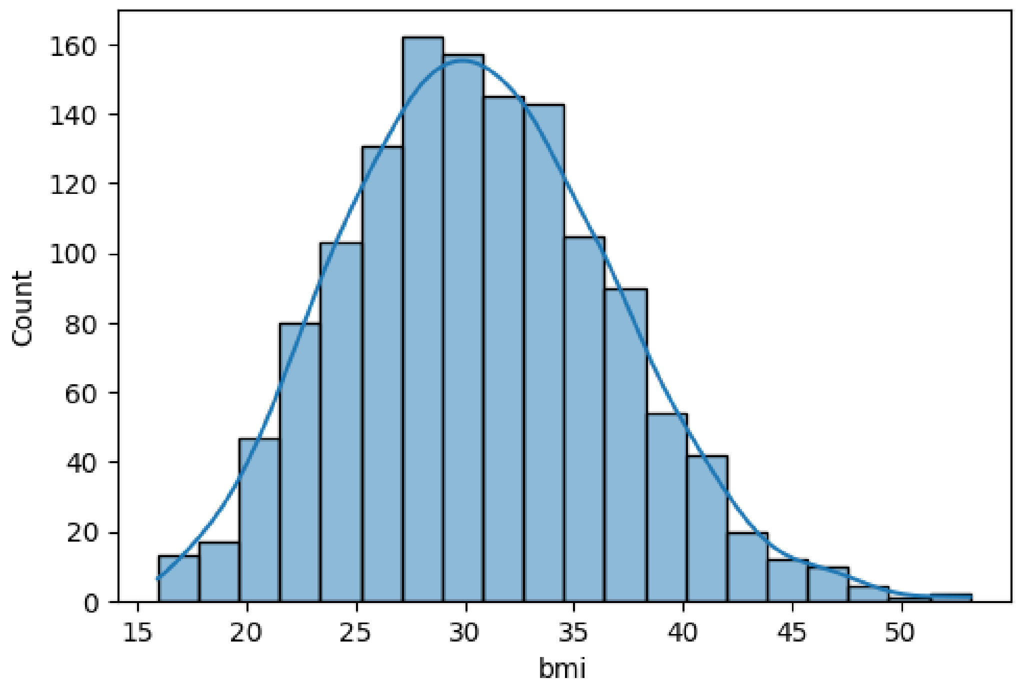
[ ]:

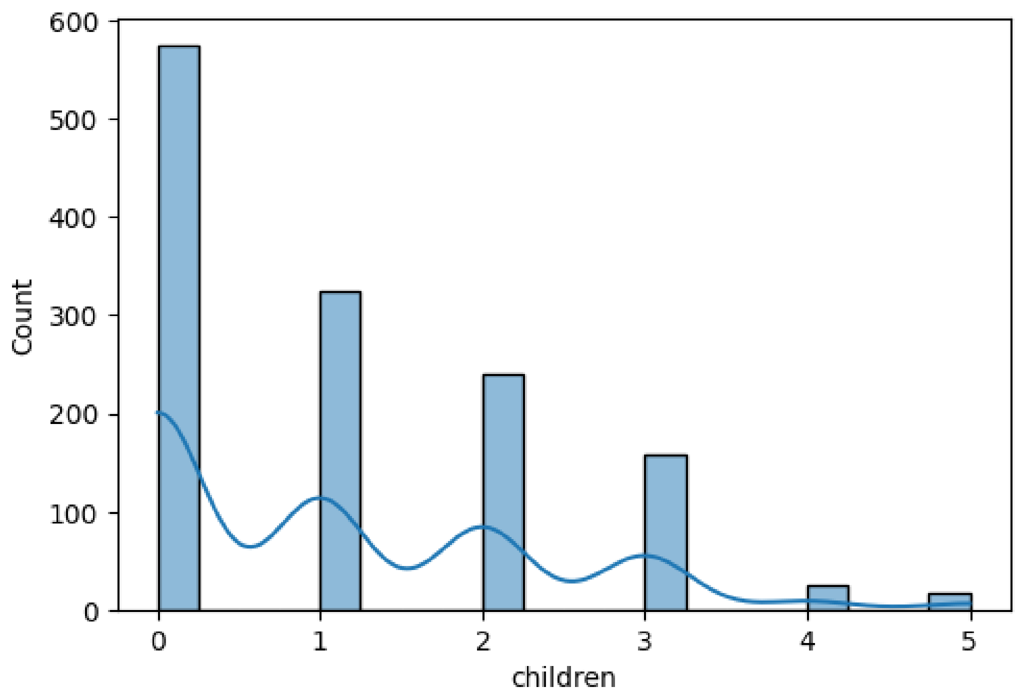
```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
```

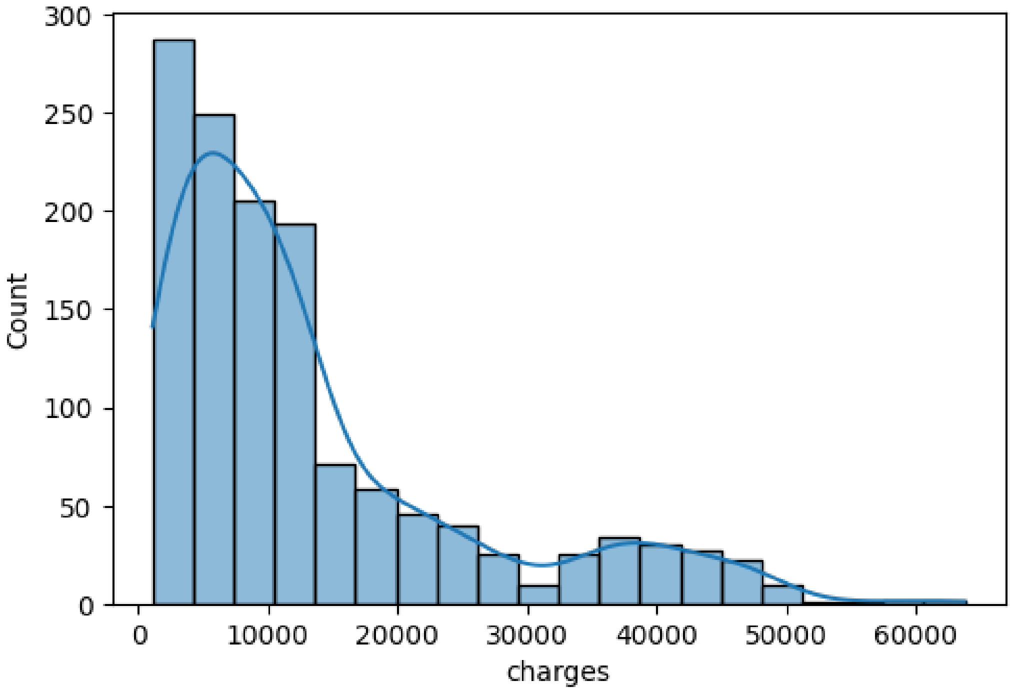
```
dtype: int64
```

[ ]:
```
numeric_columns = ['age', 'bmi', 'children', 'charges']
for col in numeric_columns:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col], bins=20, kde=True)
```



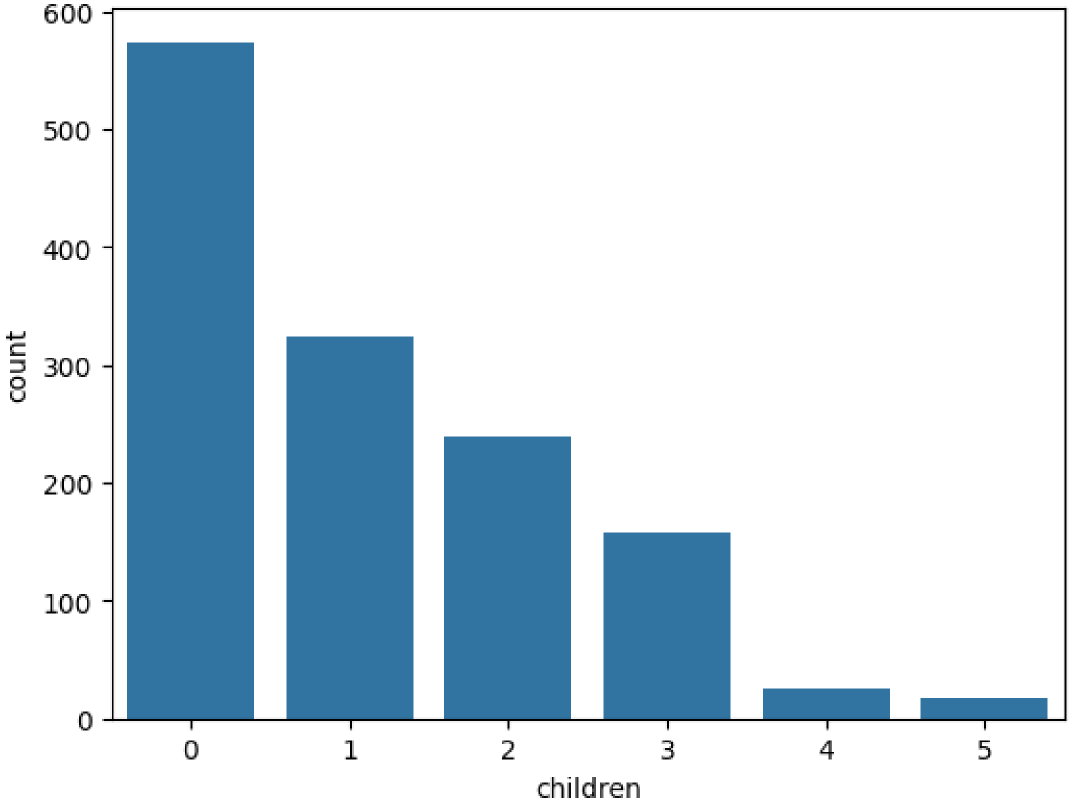

160
140
120
100
80
60
40
20
0
Count
15
20
25
30
35
40
45
50
bmi


600
500
400
300
200
100
0
Count
0
1
2
3
4
5
children



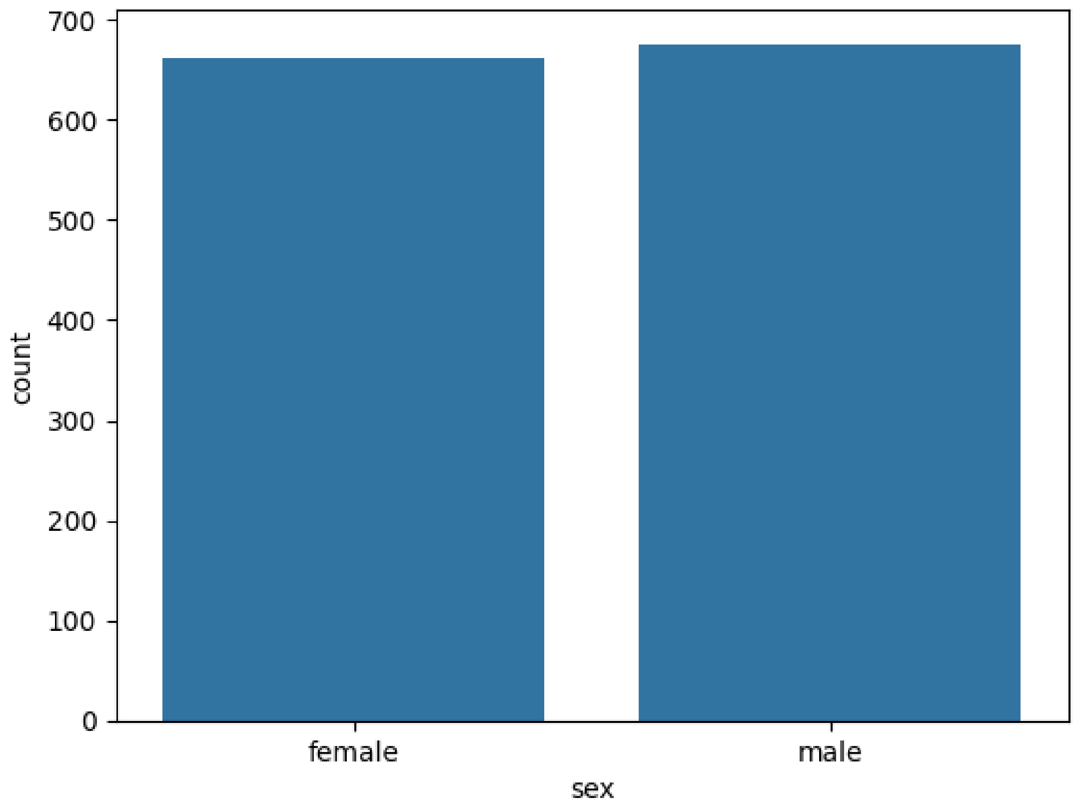
[ ]:
```
sns.countplot(x=df['children'])
```

[ ]: <Axes: xlabel='children', ylabel='count'>



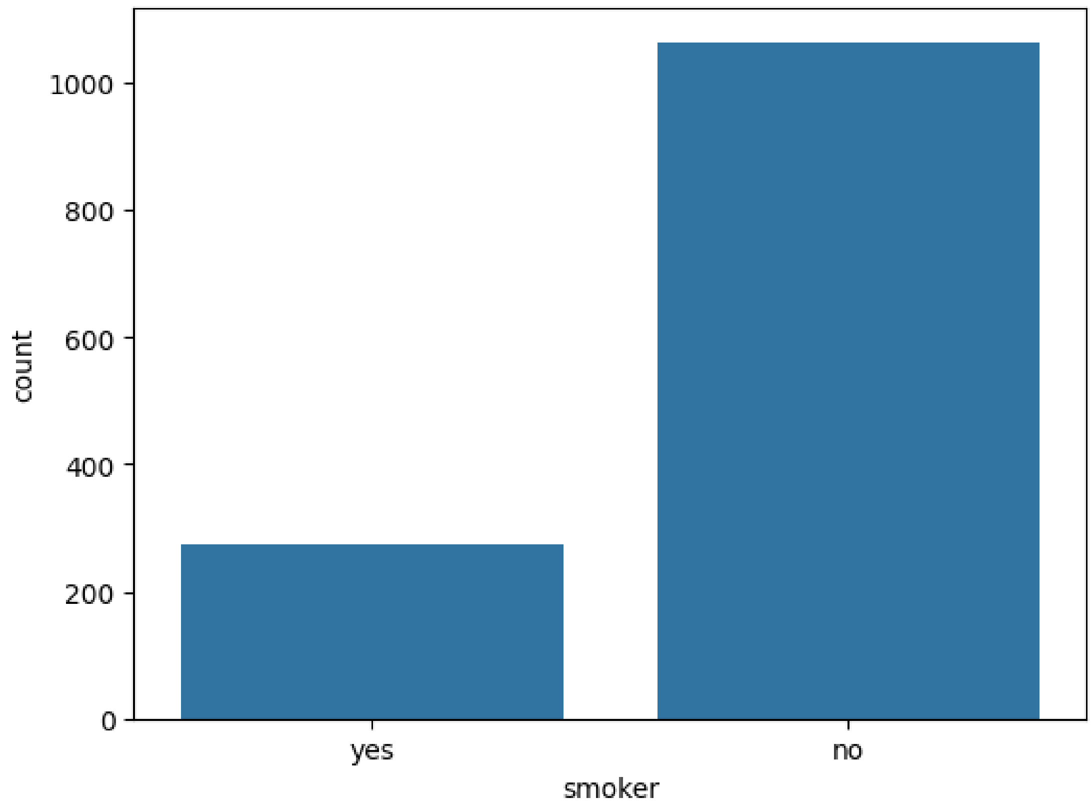
[ ]:
```
sns.countplot(x=df['sex'])
```

[ ]: <Axes: xlabel='sex', ylabel='count'>

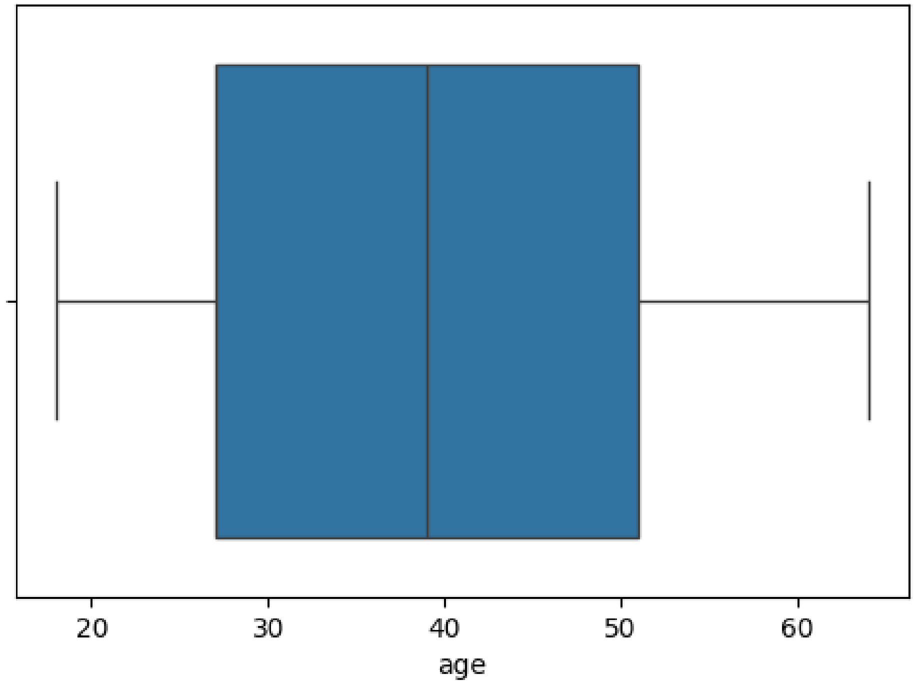


[ ]:
```
sns.countplot(x=df['smoker'])
```

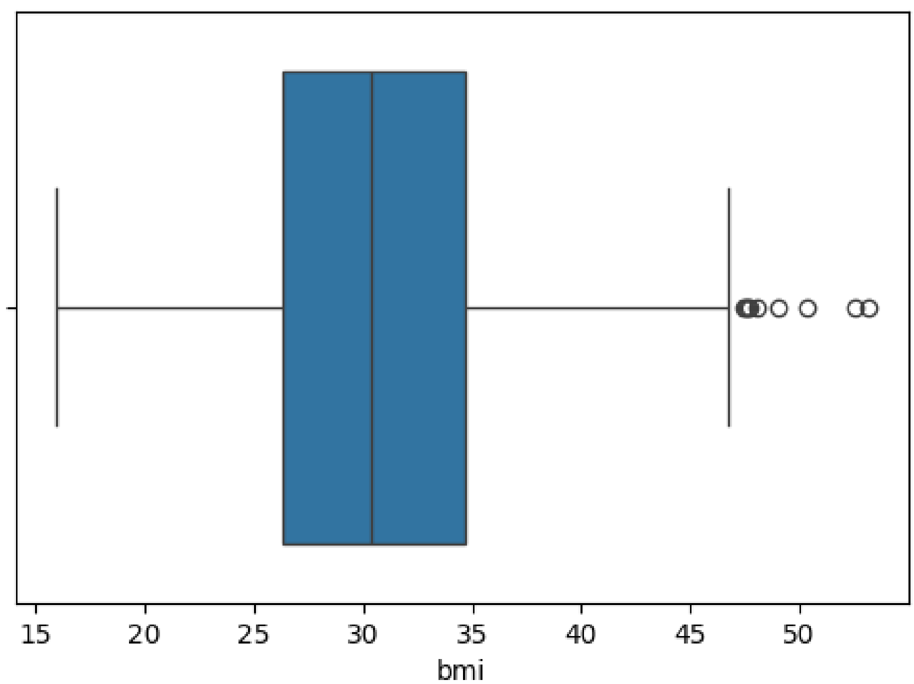
[ ]: <Axes: xlabel='smoker', ylabel='count'>



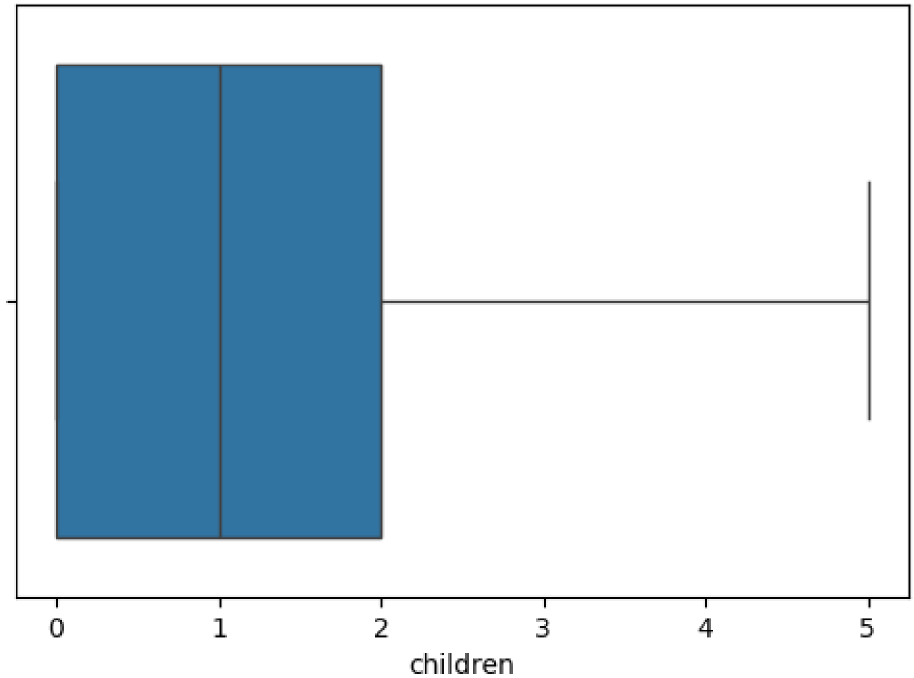
[ ]:
```
for col in numeric_columns:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col])
```

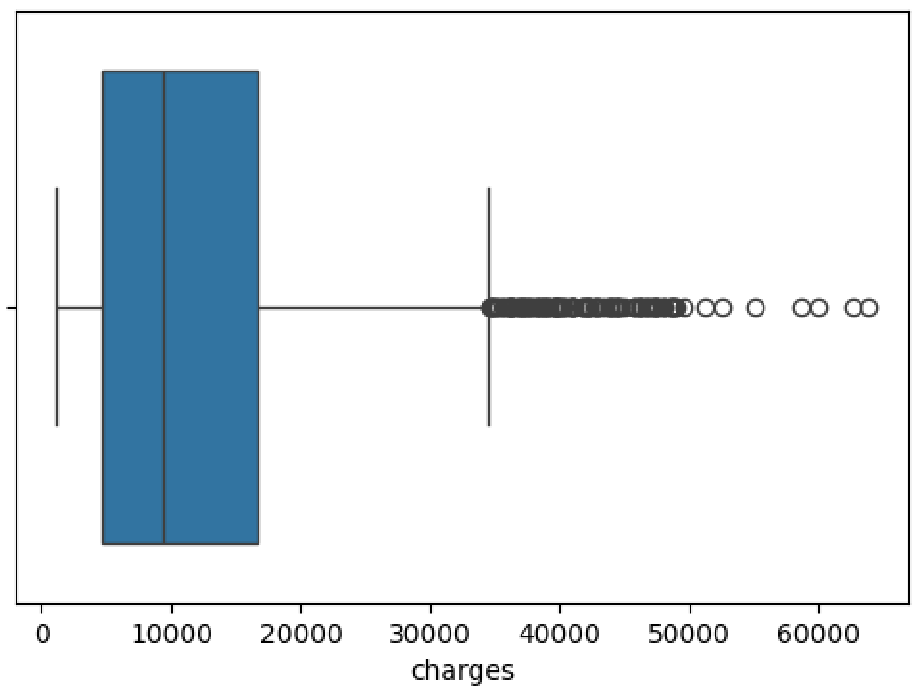


20
30
40
50
60
age



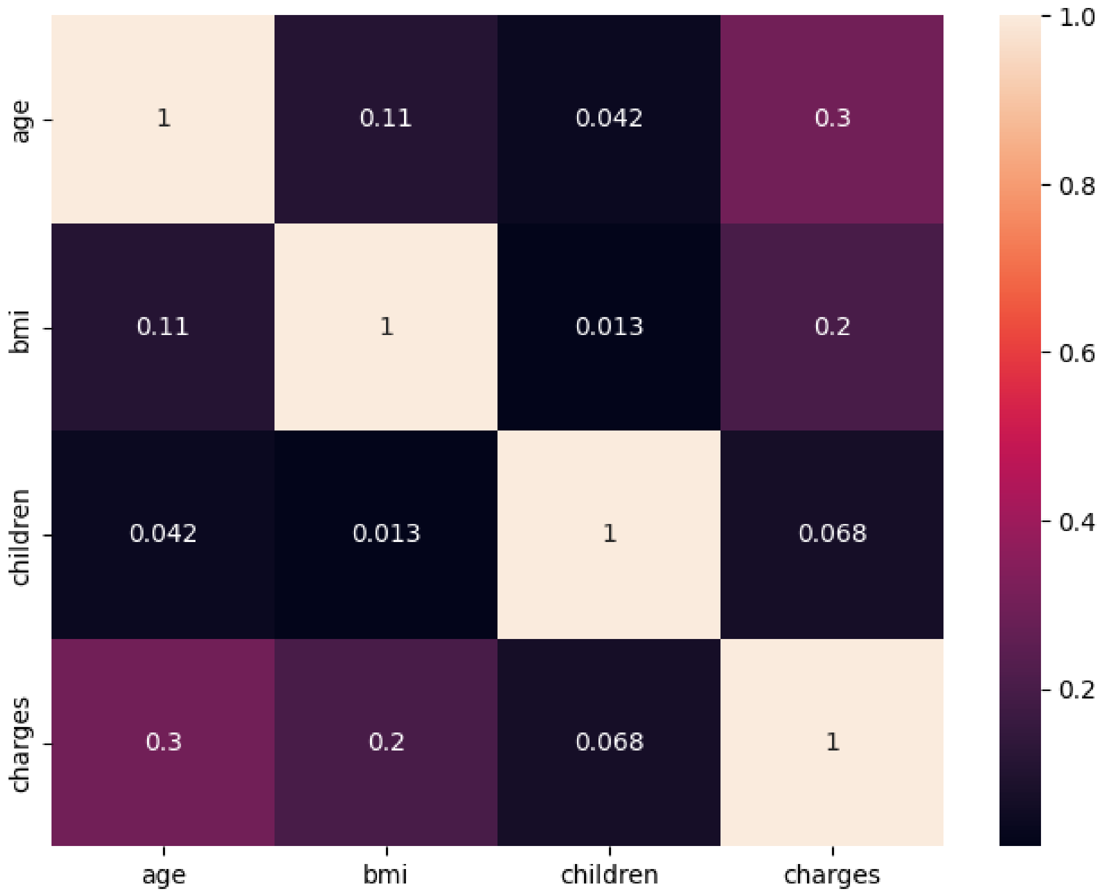
15
20
25
30
35
40
45
50
bmi


0
1
2
3
4
5
children


0
10000
20000
30000
40000
50000
60000
charges

[ ]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

[ ]: <Axes: >



2 Data Cleaning And Preprocessing

[ ]:
```
df_clean = df.copy()
```

[ ]:
```
df_clean.head()
```

[ ]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

[ ]:
```
df_clean.drop_duplicates(inplace=True)
```

[ ]:
```
df_clean.isnull().sum()
```

[ ]:
```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

[ ]:
```
df_clean.dtypes
```

[ ]:
```
age           int64
sex          object
bmi         float64
children      int64
smoker       object
region       object
charges     float64
dtype: object
```

[ ]:
```
#Labeled Incoding
df_clean['sex'] = df_clean['sex'].map({'male': 0 , 'female': 1 })
```

[ ]:
```
df_clean.head()
```

[ ]:

	age	sex	bmi	children	smoker	region	charges
0	19	1	27.900	0	yes	southwest	16884.92400
1	18	0	33.770	1	no	southeast	1725.55230
2	28	0	33.000	3	no	southeast	4449.46200
3	33	0	22.705	0	no	northwest	21984.47061
4	32	0	28.880	0	no	northwest	3866.85520

[ ]:
```
df_clean['smoker'].value_counts()
```

[ ]:
```
smoker
no     1063
yes     274
Name: count, dtype: int64
```

[ ]:
```
df_clean['smoker'] = df_clean['smoker'].map({'no': 0, 'yes': 1})
```

[ ]:
```
df_clean.head()
```

```
[ ]:    age  sex     bmi  children  smoker     region      charges
     0   19    1  27.900         0       1  southwest  16884.92400
     1   18    0  33.770         1       0  southeast   1725.55230
     2   28    0  33.000         3       0  southeast   4449.46200
     3   33    0  22.705         0       0  northwest  21984.47061
     4   32    0  28.880         0       0  northwest   3866.85520
```

```
[ ]: df_clean.rename(columns = {
         'sex':'is_female',
         'smoker':'is_smoker'
     },inplace = True)
```

```
[ ]: df['region'].value_counts()
```

```
[ ]: region
     southeast    364
     southwest    325
     northwest    325
     northeast    324
     Name: count, dtype: int64
```

```
[ ]: df_clean = pd.get_dummies(df_clean, columns=['region'], drop_first=True)
     df_clean.head()
```

```
[ ]:    age  is_female     bmi  children  is_smoker      charges  region_northwest  \
     0   19          1  27.900         0          1  16884.92400             False
     1   18          0  33.770         1          0   1725.55230             False
     2   28          0  33.000         3          0   4449.46200             False
     3   33          0  22.705         0          0  21984.47061              True
     4   32          0  28.880         0          0   3866.85520              True

        region_southeast  region_southwest
     0             False              True
     1              True             False
     2              True             False
     3             False             False
     4             False             False
```

```
[ ]: df_clean.head()
```

```
[ ]:    age  is_female     bmi  children  is_smoker      charges  region_northwest  \
     0   19          1  27.900         0          1  16884.92400             False
     1   18          0  33.770         1          0   1725.55230             False
     2   28          0  33.000         3          0   4449.46200             False
     3   33          0  22.705         0          0  21984.47061              True
     4   32          0  28.880         0          0   3866.85520              True
```

```
   region_southeast  region_southwest
0             False              True
1              True             False
2              True             False
3             False             False
4             False             False
```

[ ]:
```
df_clean = df_clean.astype(int)
```

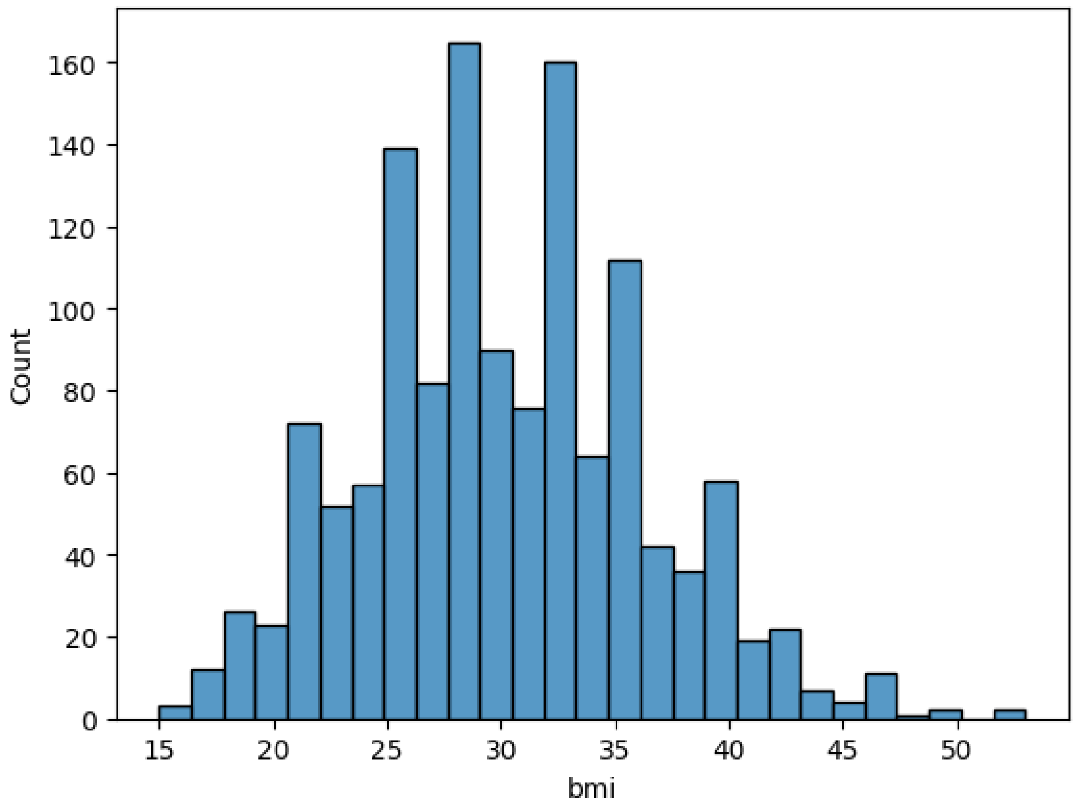
[ ]:
```
df_clean
```

[ ]:
```
       age  is_female  bmi  children  is_smoker  charges  region_northwest  \
0       19          1   27         0          1    16884                 0
1       18          0   33         1          0     1725                 0
2       28          0   33         3          0     4449                 0
3       33          0   22         0          0    21984                 1
4       32          0   28         0          0     3866                 1
…     …          …   …         …          …        …                 …
1333    50          0   30         3          0    10600                 1
1334    18          1   31         0          0     2205                 0
1335    18          1   36         0          0     1629                 0
1336    21          1   25         0          0     2007                 0
1337    61          1   29         0          1    29141                 1

      region_southeast  region_southwest
0                    0                 1
1                    1                 0
2                    1                 0
3                    0                 0
4                    0                 0
…                  …                 …
1333                 0                 0
1334                 0                 0
1335                 1                 0
1336                 0                 1
1337                 0                 0

[1337 rows x 9 columns]
```

3 Feature Enginerring and extraction

[ ]:
```
sns.histplot(df_clean['bmi'])
```

[ ]: <Axes: xlabel='bmi', ylabel='Count'>



[ ]:
```
df_clean['bmi_category'] = pd.cut(
    df_clean['bmi'],
    bins=[0, 18.5, 24.9, 29.9, float('inf')]
    , labels=['Underweight', 'Normal', 'Overweight', 'Obese']
)
```

[ ]:
```
df_clean.head()
```

[ ]:

	age	is_female	bmi	children	is_smoker	charges	region_northwest
0	19	1	27	0	1	16884	0
1	18	0	33	1	0	1725	0
2	28	0	33	3	0	4449	0
3	33	0	22	0	0	21984	1
4	32	0	28	0	0	3866	1

\

	region_southeast	region_southwest	bmi_category
0	0	1	Overweight
1	1	0	Obese
2	1	0	Obese
3	0	0	Normal
4	0	0	Overweight

[ ]:
```
df_clean['bmi_category'] = pd.cut(
    df_clean['bmi'],
    bins=[0, 18.5, 24.9, 29.9, float('inf')]
    , labels=['Underweight', 'Normal', 'Overweight', 'Obese']
)
df_clean = pd.get_dummies(df_clean, columns=['bmi_category'], drop_first=True)
```

[ ]:
```
df_clean = df_clean.astype(int)
df_clean.head()
```

[ ]:
```
   age  is_female  bmi  children  is_smoker  charges  region_northwest  \
0   19          1   27         0          1    16884                 0
1   18          0   33         1          0     1725                 0
2   28          0   33         3          0     4449                 0
3   33          0   22         0          0    21984                 1
4   32          0   28         0          0     3866                 1

   region_southeast  region_southwest  bmi_category_Normal  \
0                 0                 1                    0
1                 1                 0                    0
2                 1                 0                    0
3                 0                 0                    1
4                 0                 0                    0

   bmi_category_Overweight  bmi_category_Obese
0                        1                   0
1                        0                   1
2                        0                   1
3                        0                   0
4                        1                   0
```

[ ]:
```
df_clean.columns
```

[ ]:
```
Index(['age', 'is_female', 'bmi', 'children', 'is_smoker', 'charges',
       'region_northwest', 'region_southeast', 'region_southwest',
       'bmi_category_Normal', 'bmi_category_Overweight', 'bmi_category_Obese'],
      dtype='object')
```

[ ]:
```
#Feature Scaling  - transform the numbers b/w 0 and 1
from sklearn.preprocessing import StandardScaler
cols = ['age', 'bmi', 'children']
scaler = StandardScaler()
df_clean[cols] = scaler.fit_transform(df_clean[cols])
```

[ ]:
```
df_clean.head()
```

```
[ ]:        age  is_female       bmi  children  is_smoker  charges  \
     0 -1.440418          1 -0.517949 -0.909234          1    16884
     1 -1.511647          0  0.462463 -0.079442          0     1725
     2 -0.799350          0  0.462463  1.580143          0     4449
     3 -0.443201          0 -1.334960 -0.909234          0    21984
     4 -0.514431          0 -0.354547 -0.909234          0     3866

        region_northwest  region_southeast  region_southwest  bmi_category_Normal  \
     0                 0                 0                 1                    0
     1                 0                 1                 0                    0
     2                 0                 1                 0                    0
     3                 1                 0                 0                    1
     4                 1                 0                 0                    0

        bmi_category_Overweight  bmi_category_Obese
     0                        1                   0
     1                        0                   1
     2                        0                   1
     3                        0                   0
     4                        1                   0
```

```
[ ]: #Feature extraction is a process of selecting the right feature
```

```
[ ]: from scipy.stats import pearsonr

     #---------------------------------
     # Pearson correlation calculation
     #---------------------------------

     #List of feature to check against target 'charges'
     selected_features = [
         'age', 'bmi', 'children', 'is_female', 'is_smoker',
         'region_northwest', 'region_southeast', 'region_southwest',
         'bmi_category_Normal', 'bmi_category_Overweight', 'bmi_category_Obese'
     ]

     correlations = {
         feature : pearsonr(df_clean[feature], df_clean['charges'])[0]
         for feature in selected_features
     }
     correlation_df = pd.DataFrame(list(correlations.items()), columns=['Feature',
      'Correlation'])
     correlation_df.sort_values(by='Correlation', ascending=False)
```

```
[ ]:                   Feature  Correlation
     4               is_smoker     0.787234
     0                     age     0.298309
```

```
10       bmi_category_Obese     0.200348
1                       bmi     0.196236
6          region_southeast     0.073577
2                  children     0.067390
5          region_northwest    -0.038695
7          region_southwest    -0.043637
3                 is_female    -0.058046
8       bmi_category_Normal    -0.104042
9   bmi_category_Overweight    -0.120601
```

[ ]:
```
#chi square test
# When both features (X) and the target (Y) are categorical, we use the␣
↪Chi-Square test
# to check if a feature and the target are independent or not.

cat_feature = [
    'is_female', 'is_smoker',
    'region_northwest', 'region_southeast', 'region_southwest',
    'bmi_category_Normal', 'bmi_category_Overweight', 'bmi_category_Obese'
]
```

[ ]:
```
from scipy.stats import chi2_contingency
import pandas as pd

alpha = 0.05

df_clean['charges_bin'] = pd.qcut(df_clean['charges'], q=4, labels=False)
chi2_results = {}

for col in cat_feature :
    contingency = pd.crosstab(df_clean[col], df_clean['charges_bin'])
    chi2_stat, p_val, _, _ = chi2_contingency(contingency)
    decision = 'Reject Null (Keep Feature)' if p_val < alpha else 'Accept Null␣
↪(Drop Feature)'
    chi2_results[col] = {
        'chi2_statistic': chi2_stat,
        'p_value': p_val,
        'Decision': decision
    }

chi2_df = pd.DataFrame(chi2_results).T
chi2_df = chi2_df.sort_values(by='p_value')
chi2_df
```

[ ]:
```
                           chi2_statistic   p_value                    Decision
is_smoker                      848.219178       0.0  Reject Null (Keep Feature)
region_southeast                15.998167  0.001135  Reject Null (Keep Feature)
```

```
is_female                    10.258784   0.01649  Reject Null (Keep Feature)
bmi_category_Obese            8.515711  0.036473  Reject Null (Keep Feature)
region_southwest              5.091893  0.165191  Accept Null (Drop Feature)
bmi_category_Overweight        4.25149  0.235557  Accept Null (Drop Feature)
bmi_category_Normal           3.708088   0.29476  Accept Null (Drop Feature)
region_northwest               1.13424  0.768815  Accept Null (Drop Feature)
```

[ ]:
```
final_df = df_clean[['age', 'is_female', 'bmi', 'children', 'is_smoker',␣
 ↪'charges','region_southeast','bmi_category_Obese']]
```

[ ]:
```
final_df
```

[ ]:
```
           age  is_female       bmi  children  is_smoker  charges  \
0    -1.440418          1 -0.517949 -0.909234          1    16884
1    -1.511647          0  0.462463 -0.079442          0     1725
2    -0.799350          0  0.462463  1.580143          0     4449
3    -0.443201          0 -1.334960 -0.909234          0    21984
4    -0.514431          0 -0.354547 -0.909234          0     3866
…          …        …         …         …          …        …
1333  0.767704          0 -0.027743  1.580143          0    10600
1334 -1.511647          1  0.135659 -0.909234          0     2205
1335 -1.511647          1  0.952670 -0.909234          0     1629
1336 -1.297958          1 -0.844753 -0.909234          0     2007
1337  1.551231          1 -0.191145 -0.909234          1    29141

      region_southeast  bmi_category_Obese
0                    0                   0
1                    1                   1
2                    1                   1
3                    0                   0
4                    0                   0
…                  …                   …
1333                 0                   1
1334                 0                   1
1335                 1                   1
1336                 0                   0
1337                 0                   0

[1337 rows x 8 columns]
```

[ ]:
```
df
```

[ ]:
```
      age     sex     bmi  children smoker     region      charges
0      19  female  27.900         0    yes  southwest  16884.92400
1      18    male  33.770         1     no  southeast   1725.55230
2      28    male  33.000         3     no  southeast   4449.46200
3      33    male  22.705         0     no  northwest  21984.47061
```

```
4       32    male  28.880         0     no  northwest   3866.85520
…    …      …      …        …     …        …          …
1333    50    male  30.970         3     no  northwest  10600.54830
1334    18  female  31.920         0     no  northeast   2205.98080
1335    18  female  36.850         0     no  southeast   1629.83350
1336    21  female  25.800         0     no  southwest   2007.94500
1337    61  female  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]
```

[ ]: